



Fig. 6: Working of a passive RFID tag
(Source: Link)

mon energy harvesting ICs from e-peas.

Radio frequency identification (RFID)

The passive RFID tags do not have any power source, they harness the energy from the radiated electromagnetic waves from the RFID readers. The coiled antenna within the tag forms a magnetic field and draws power to run the IC. The tag then sends the information over to the reader. The working of an RFID is shown in Fig. 6.

Near Field Communication (NFC)

NFC is like RFID but works at a maximum distance of 10cm and on a frequency of 10.56MHz. It is a secure mode of data sharing and provides peer-to-peer sharing. NFC is used for various purposes, but one of the most common purposes these days is in contactless debit and credit cards for transactions. Its application is shown in the box below.

Where we can use an energy harvester

The IoT industry is growing at a compound annual global rate of 26.9% and the global IoT market size is expected to exceed US\$870 billion by 2025. With the adoption of IoT and an increase in data sensing and collection, standalone systems make sense. Some of the applications of energy harvesting are:

- Wearable fitness
- Medical devices

Contactless payment using NFC

1. A contactless card contains a chip and antenna; the chip contains your secure information while the antenna works to power the chip as well as to transfer the data.
2. The reader emits an electromagnetic (EM) signal.
3. When a card is placed in the vicinity of EM wave, the waves power the chip and a communication between the chip and the reader is initiated.
4. When the reader recognises and verifies the data, a secure communication is formed and encrypted data transfer takes place between the devices.
5. The card reader confirms the payment.



Fig. 7: Contactless payment using NFC (Source: Link)

TABLE 5
COMPARISON OF ENERGY HARVESTING DEVICES

Model	Manufacturer	Energy storage type	Energy it can harness	Output voltage (max)	MPPT
AEM40940	e-peas semiconductor	Rechargeable batteries, Capacitors	RF energy	5V	Yes
AEM40940	Powercast	Rechargeable batteries, Capacitors	RF energy	4.3V	Yes
AEM40940	RF Diagnostics	Rechargeable batteries, Capacitors	RF energy	4.2V	Yes
AEM40940	Maxim Integrated	Rechargeable batteries, Capacitors	Solar	4.7V	Yes
AEM40940	STM	Rechargeable batteries, Capacitors	Thermal and solar	5.3V	Yes
AEM40940	e-peas semiconductor	Rechargeable batteries, Capacitors	Thermal energy	4.5V	Yes

- Industrial IoT sensors
- Asset tracking devices
- Wireless sensor networks
- Smart homes
- Smart agriculture
- Weather forecasting
- Smart cities
- Automobiles

The future of energy harvesting

The advancements in electronics industry pave the way for higher-efficiency components that will

increase the adoption of energy harvesting techniques. Till recent years, most of the focus was on transducers which essentially made harnessing possible, but now companies such as Texas Instruments, e-peas, STM, etc are working on improving the energy harvester ICs. Some companies have started integrating ICs with on-chip energy harvesting solutions. Devices such as Atmosic ATM3 series Bluetooth SoC solutions have on-chip RF harvesting.